Acute Stabilization Pathway Tachyarrhythmia: Outpatient Pathway

PEDIATRIC CARE

Background:

KP Medical Office Buildings principally function as doctors' offices set up for rapid turn-around of outpatient encounters. As such, they are not designed or well equipped for patient observation or prolonged management. MOBs, staffed by medical assistants and licensed vocational nurses, are typically not staffed robustly enough to allow for efficient outpatient clinic workflow (ie, rooming patients, taking vitals, preparing for outpatient encounters) simultaneous with the continuous direct observation of one patient needing extra attention. Furthermore, there are no complete crash carts in these facilities, and electrocardiograms may or may not be available without transfer to another building. Skill and comfort to obtain intravenous access may be limited, and staffing limitations may make it difficult for more than one practitioner to assist with IV access. Patients with tachyarrhythmia in the MOB setting may rapidly clinically deteriorate with few options to intervene. Due to these factors, it may be difficult or even unsafe to care for a child in an MOB with any unstable or potentially unstable tachyarrhythmia, while awaiting transfer to an emergency department or inpatient setting.

Defining Tachyarrhythmia:

Tachycardia may be a sign of a primary arrythmia, or a cardiac output issue such as shock or sepsis. In infants and toddlers who are often frightened at the MOB, it may be difficult to differentiate between the various etiologies of tachycardia. Heart rates above 200/minute are very likely to be a primary arrhythmia in any aged patient. Above 160/min in a child over age 5 is likely to be a primary tachycardia.

Primary tachycardias are usually exactly the same fast heart rate without any variation. Patients in whom the heart rate is elevated out of proportion to their appearance are more likely to have had a primary tachycardia.

Secondary sinus tachycardias are typically fast, but variable, and respond to treatment of an associated condition, such as lysing fever or giving volume. Patients with fever, or other signs of systemic illness such as cough, wheezing, rash, vomiting, diarrhea, or seizures are more likely to have a secondary, sinus tachycardia.

Determining whether a tachycardia is a primary arrhythmia or a secondary sinus tachycardia requires an electrocardiogram and a competent reader. If there is an ECG machine in the unit, it is almost always reasonable to perform an ECG if primary arrhythmia is in the differential, unless the patient is so unstable that the ECG would not change the management of the patient (cold, pulseless, somnolent, ineffective respirations). These will be rare occurrences. If the ECG machine is in a different unit, patients should, in general, not be sent unattended to the other unit for the ECG. Exceptions include: patients who are otherwise asymptomatic and whose chief complaint is the tachycardia itself (especially ambulatory, verbal patients), and infants in whom tachycardia is found by serendipity by the examining physician while the infant presented for an unrelated chief complaint or for a well-child visit.

<5 years, Heart Rate > 200 | > 5 years, Heart Rate >160

Presents with any of the following signs/symptoms of poor cardiac output:

Minor signs/symptoms

- Decreased Urine Output
- Dizziness/Lightheadedness (older child)
- Poor Feeding (infant)
- Pallor
- Cool Skin

MAJOR signs/symptoms

- Chest Discomfort (older child)
- Lethargy/Somnolence
- Cyanosis
- Delayed Capillary Refill
- Hypotension
- Poor pulses
- Ineffective Respirations

Yes – Patient In Shock

Immediate Resuscitation/Intervention

- Maintain Airway – sniff position.
- 100% oxygen – face mask/non-rebreather.
- Place on cardiac monitor if available.

Local ER on site?

NO → **Call 9-1-1 immediately** to activate EMS transfer of unstable patient to nearest Emergency Facility.

Yes →

- **Immediately transfer patient to ED accompanied by medical staff.**
- If readily available in clinic, may attempt vagal maneuvers (Infant: ice to face. Older child: Valsalva – bearing down, coughing, blowing into a syringe).
- Call PTAC to notify Pediatric Cardiology of patient with unstable arrhythmia.

No – Patient Currently Stable

May attempt to determine 1^{o} vs. 2^{o} tachycardia

- Place patient on monitor if immediately available.
- Perform 12-lead ECG if immediately available.
- While ECG taking place, physician to contact Peds Cardiologist on-call.
- Transmit ECG to Cardiologist for evaluation.
- If unable to reach Cardiologist via SCPMG-On-Call or Cortext, contact PTAC to assist.
- Determination of primary vs. secondary arrhythmia should be confirmed by cardiologist.

Primary Arrhythmia?

No → Continue further workup. Attempt to alleviate primary reason for tachycardia (ie. treat fever/pain, calm patient, hydrate patient, etc.).

Yes →

- **Consider transfer to ED if on site, given potential for rapid clinical deterioration.**
- **While arranging transport to higher level of care (ED, inpatient),** may attempt vagal maneuvers (Infant: ice to face. Older child: Valsalva – bearing down, coughing, blowing into a syringe).
- If staff is competent and comfortable with intravenous access, and medications are available for resuscitation (i.e. at an infusion center or branded urgent care with crash cart and defibrillator), may attempt to place PIV (preferable in antecubital area).
- Adenosine for arrest of SVT should only be attempted if a defibrillator and resuscitation medications are immediately available and ready for use by competent PALS certified staff.
- If IV start is difficult, or no resuscitation medications available onsite, arrange immediate transport while simultaneously attempting vagal maneuvers.

If at any point during the encounter the treating clinician feels that the patient has become unstable or unsafe, **transfer to local ER or call 9-1-1 immediately.**